SECTION A – 1 MARK QUESTIONS (10 × 1 = 10 Marks)

1. What is a hallucination in LLMs?

   a) Model overconfidence
   b) Generating plausible but false information
   c) Processing error
   d) Slow inference

2. Which mitigation strategy directly addresses hallucinations?

   a) Larger models
   b) RAG architecture
   c) More training data
   d) Faster inference

3. What is AI bias?

   a) Model's preference for certain tasks
   b) Systematic errors favoring/disadvantaging groups unfairly
   c) Random noise
   d) Computational preference

4. How is fairness measured in AI systems?

   a) Model accuracy only
   b) Metrics like demographic parity, equalized odds, calibration
   c) Training time
   d) Model size

5. What is prompt injection?

   a) Adding more tokens
   b) Embedding keywords
   c) Malicious instructions embedded in user input
   d) Data compression

6. How can prompt injection be prevented?

   a) Increase model size
   b) Input validation, hardened system prompts, output filtering
   c) Use larger vocabulary
   d) Train longer

7. What is the primary goal of content moderation in AI?

a) Reduce inference speed
b) Filter harmful, inappropriate, or dangerous content
c) Increase model parameters
d) Improve training spee

8. What is GDPR in the context of AI?

a) Graphics Data Processing Regulation
b) EU data protection law requiring consent and privacy
c) General AI Data Requirements
d) Global Data Performance Rule

9. How should PII (Personally Identifiable Information) be handled?

a) Store openly in logs
b) Encrypt, mask, and minimize collection
c) Include in model training
d) Share with third parties

10. What is a fairness audit?

a) Financial audit of AI company
b) Comprehensive evaluation of model bias and discrimination
c) Security check
d) Performance test

SECTION B – 5 MARK QUESTIONS Answer ANY 6 (6 × 5 = 30 Marks)

1. Define Generative AI with examples.

2. Explain zero-shot prompting.

3. What is BLEU score?

4. Define prompt injection.

5. Explain API integration in AI.

6. Write short notes on Generative Adversarial Networks (GANs).

7. What is semantic similarity in AI?

SECTION C – 10 MARK QUESTIONS Answer ANY 4 (4 × 10 = 40 Marks)

1. Explain full AI model deployment lifecycle.

2. Discuss AI security threats and mitigation.

3. Explain embeddings and vector databases in detail.

4. Explain multimodal AI systems with examples.

5. Discuss AI bias and fairness in model training.

SECTION C – 20 MARK CASE STUDY Answer ANY 1 (1 × 20 = 20 Marks)

CASE STUDY 1: MANUFACTURING QUALITY CONTROL AI

SCENARIO:

ManufacturePro deploys computer vision and generative AI for real-time quality inspection on assembly lines. The system is expected to catch defects. Issues:

AI misses critical defects, allowing faulty products to reach customers

Model accuracy varies by product type and manufacturing line

Data drift: Model performance degrades as line conditions change

False positives create production bottlenecks and waste

No explanation for defect decisions affecting operator trust

QUESTIONS

Q1 Design evaluation metrics balancing sensitivity (catching defects) vs. specificity (avoiding false alarms) with acceptable failure thresholds.

Q2 Implement continuous monitoring detecting model drift and automating retraining when performance degrades below thresholds.

Q3 Create explainability system showing operators which image features flagged defects improving trust and debugging.

Q4 Design human-in-the-loop system for high-uncertainty predictions with quality metrics per production line and time.

CASE STUDY 2: PERSONALIZED MENTAL HEALTH AI CHATBOT

SCENARIO:

MindWell launches an AI chatbot providing mental health support and crisis intervention. The system uses LLMs fine-tuned on therapy conversations. Issues:

AI provides harmful advice to suicidal individuals

System lacks understanding of crisis severity and urgency

Privacy breaches: Sensitive mental health data exposure

Inadequate escalation to human professionals

Liability concerns and regulatory non-compliance (HIPAA, FDA)

QUESTIONS

Q1 Develop crisis detection system identifying suicide risk and immediately escalating to human counselors with documented protocols.

Q2 Design comprehensive safety framework including guardrails, training on therapeutic ethics, and harmful response filtering.

Q3 Create privacy-first architecture with end-to-end encryption and HIPAA compliance for sensitive mental health information.

Q4 Establish FDA regulatory compliance and liability frameworks with informed consent, professional oversight, and incident response procedures